HAN CHIANG HIGH SCHOOL, PENANG
TRIAL EXAMINATION 2026
ADVANCED MATHEMATICS II
(SE07)

PAPER 2 SUBJECTIVE QUESTIONS

Class : S3
Duration : 120 minutes

INSTRUCTIONS TO CANDIDATES

1. This subject comprises of two papers:
   Paper 1 : Multiple-choice question (30%)
   Paper 2 : Subjective question (70%)
2. Paper 2 consists of :
   Question 1 to question 5: 4% for each question
   Question 6 to question 10: 6% for each question
   Question 11 to question 12: 10% for each question
   You are required to attempt all **twelve** questions.
3. Begin each question on a fresh page.
4. Use only **blue** pen or **black** pen to write your answers and use pencil for drawing only.
5. Do not copy the questions, but the question numbers should be clearly written.
6. Show all mathematical working clearly. Geometric figures should be drawn where necessary.
7. Unless otherwise specified, the prescribed electronic calculators may be used.
8. Arrange the answer scripts in numerical order and tie them together.
9. This paper consists of 6 printed pages.

DO NOT TURN OVER THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

Student's Name: _____ Student's Number: _____

Advanced Mathematics (II) Formula Sheet

I. Algebra

$a^3 \pm b^3 = (a \pm b)(a^2 \mp ab + b^2)$

$\log_a(xy) = \log_a x + \log_a y$

$\log_a \frac{x}{y} = \log_a x - \log_a y$

$\log_a x^m = m \log_a x$

$a^{\log_a x} = x$

$\log_a x = \frac{\log_b x}{\log_b a}$

$(a+b)^n = \sum_{r=0}^{n} {}^nC_r \, a^{n-r} b^r$

$A^{-1} = \frac{1}{|A|} \operatorname{adj} A$

$\frac{a_1 + a_2 + \cdots + a_n}{n} \geq \sqrt[n]{a_1 a_2 \cdots a_n}$

$\left[r(\cos\theta + i\sin\theta)\right]^n = r^n(\cos n\theta + i \sin n\theta)$

A.P. $a_n = a + (n-1)d$

$S_n = \frac{n}{2}\left[2a + (n-1)d\right]$

G.P. $a_n = ar^{n-1}$

$S_n = \frac{a(1-r^n)}{1-r}$

$S_\infty = \frac{a}{1-r}$

$\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$

$\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$

$\sum_{k=1}^{n} k^3 = \left[\frac{n(n+1)}{2}\right]^2$

Accumulated value after n years $= p \times \left(1 + \frac{\frac{r}{m}}{100}\right)^{nm}$

$(1+x)^n = 1 + \frac{n}{1!}x + \frac{n(n-1)}{2!}x^2 + \cdots + \frac{n(n-1)\cdots(n-r+1)}{r!}x^r + \cdots$

II. Trigonometry

Arc length $= \theta r$

Area of sector $= \frac{1}{2}\theta r^2$

$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$

$a^2 = b^2 + c^2 - 2bc\cos A$

Area of triangle $= \frac{1}{2}ab\sin C$

Area of triangle $= \sqrt{s(s-a)(s-b)(s-c)}$,

$s = \frac{a+b+c}{2}$

$\sin^2\theta + \cos^2\theta = 1$

$1 + \tan^2\theta = \sec^2\theta$

$1 + \cot^2\theta = \operatorname{cosec}^2\theta$

$\sin(A \pm B) = \sin A\cos B \pm \cos A \sin B$

$\cos(A \pm B) = \cos A\cos B \mp \sin A \sin B$

$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$

$\sin 2A = 2\sin A\cos A$

$\cos 2A = \cos^2 A - \sin^2 A$

$= 2\cos^2 A - 1$

$= 1 - 2\sin^2 A$

$\tan 2A = \frac{2\tan A}{1 - \tan^2 A}$

$\sin A\cos B = \frac{\sin(A+B) + \sin(A-B)}{2}$

$\cos A\cos B = \frac{\cos(A+B) + \cos(A-B)}{2}$

$\sin A\sin B = \frac{\cos(A-B) - \cos(A+B)}{2}$

III. Analytic Geometry

Distance from a point to a line $d = \dfrac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$

Division of a line segment $\left(\dfrac{mx_2 + nx_1}{m+n}, \dfrac{my_2 + ny_1}{m+n} \right)$

Area of triangle $= \dfrac{1}{2} \left| (x_1y_2 + x_2y_3 + x_3y_1) - (x_2y_1 + x_3y_2 + x_1y_3) \right|$

Equation of a line $y - y_1 = m(x - x_1)$

Standard equation of a circle $(x-h)^2 + (y-k)^2 = r^2$

Translation $\begin{cases} x = x' + h \\ y = y' + k \end{cases}$

Rotation of axes $\begin{cases} x = x'\cos\theta - y'\sin\theta \\ y = x'\sin\theta + y'\cos\theta \end{cases}$

Parabola

Standard equation	$y^2 = 4ax$
Focus	$(a, 0)$
Directrix	$x + a = 0$

Ellipse

Standard equation	$\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$
Eccentricity	$e = \dfrac{\sqrt{a^2 - b^2}}{a}$
Foci	$(\pm ae, 0)$
Directrices	$x \pm \dfrac{a}{e} = 0$

Hyperbola

Standard equation	$\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 1$
Eccentricity	$e = \dfrac{\sqrt{a^2 + b^2}}{a}$
Foci	$(\pm ae, 0)$
Directrices	$x \pm \dfrac{a}{e} = 0$
Asymptotes	$y = \pm \dfrac{b}{a} x$

Inner product $\underline{a} \cdot \underline{b} = |\underline{a}||\underline{b}|\cos\theta$

Magnitude of the cross product $|\underline{a} \times \underline{b}| = |\underline{a}||\underline{b}|\sin\theta$

Projection of vector $\underline{a}$ onto vector $\underline{u}$ $\quad \text{proj}_{\underline{u}}\,\underline{a} = \dfrac{\underline{a} \cdot \underline{u}}{|\underline{u}|^2}\underline{u}$

IV. Statistics and Probability

Mean $\bar{x} = \dfrac{\sum_{i=1}^{n} x_i}{n}$

Mode $= L + \left(\dfrac{d_1}{d_1 + d_2} \right) C$

Median $m = L_m + \left(\dfrac{\frac{n}{2} - F_m}{f_m} \right) C_m$

Upper quartile $Q_3 = L_3 + \left(\dfrac{\frac{3n}{4} - F_3}{f_3} \right) C_3$

Lower quartile $Q_1 = L_1 + \left(\dfrac{\frac{n}{4} - F_1}{f_1} \right) C_1$

Interquartile range $IQR = Q_3 - Q_1$

Quartile deviation $Q.D. = \dfrac{Q_3 - Q_1}{2}$

Variance $\sigma^2 = \dfrac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n} = \dfrac{\sum_{i=1}^{n} x_i^2}{n} - \bar{x}^2$

Covariance $\sigma_{XY} = \dfrac{1}{n} \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})$

$= \dfrac{1}{n} \sum_{i=1}^{n} (x_i y_i) - \bar{x} \cdot \bar{y}$

Coefficient of correlation $r = \dfrac{\sigma_{XY}}{\sigma_X \sigma_Y}$

Equation of the regression line

$\hat{y} - \bar{y} = \dfrac{\sigma_{XY}}{\sigma_X^2} (x - \bar{x})$

Statistical index $I = \dfrac{Q_1}{Q_0} \times 100$

Composite index $= \dfrac{\sum_{i=1}^{n} w_i x_i}{\sum_{i=1}^{n} w_i}$

${}^{n}P_r = \dfrac{n!}{(n-r)!}$

${}^{n}C_r = \dfrac{n!}{(n-r)!r!}$

$P(E_k \mid A) = \dfrac{P(A \cap E_k)}{\sum_{i=1}^{n} P(A \mid E_i) P(E_i)}$

Random variable X

$E(X) = x_1 p_1 + x_2 p_2 + \cdots + x_k p_k$

$\sigma_X^2 = E(X^2) - (E(X))^2$

Binomial distribution $P(X = r) = {}^{n}C_r \, p^r (1-p)^{n-r}$

$\mu = np$

$\sigma = \sqrt{np(1-p)}$

Normal distribution $Z = \dfrac{X - \mu}{\sigma}$

Answer all questions.

1. a) Evaluate $\int_0^{\pi} x\sin 2x \, dx$ [2%]

b) Let $y=2\sin\theta$, evaluate $\int_{-2}^{2} \sqrt{4-y^2} \, dy$. [2%]

2. a) Find a nonzero vector that is perpendicular/orthogonal to the plane that passes through points P(3,1,1), Q(5,2,4) and R(8,5,3). [3%]

b) Hence, find the area of triangle PQR. [1%]

3. Solve the equation $z^3 - 4\sqrt{3} - 4i = 0$, giving your solution in form of $r(\cos\theta + i\sin\theta)$, where \\

$-\pi < \theta \leq \pi$. [4%]

4. Diagram 1 shows part of the curve with equation $y = e^{0.5x^2}$.

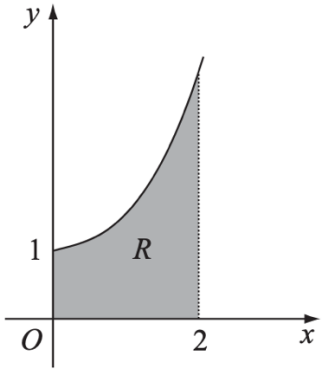


Diagram 1

The finite region R, shown shaded in diagram 1, is bounded by the curve, x-axis, y-axis and line $x=2$.

a) Complete the table. [2%]

x	**0**	**0.4**	**0.8**	**1.2**	**1.6**	**2.0**
y		$e^{0.08}$				

b) Use the trapezium rule with all the values in the table to find an approximate value for the area of R, giving your answer to 4 significant figures. [2%]

5. If $z = \cos\theta + i\sin\theta$, prove that $\cos^4\theta = \frac{1}{8}\cos 4\theta + \frac{1}{2}\cos 2\theta + \frac{3}{8}$. [4%]

6. a) Find the foci of the ellipse $25(x+1)^2 + 9(y+2)^2 = 225$. [3%]

b) State the eccentricity of the curve represented by equation $x^2 + 2y^2 - 2x + 3y + 2 = 0$. [3%]

7. a) Solve $2 \cdot 3^{2x+2} + 19 \cdot 6^x - 12 \cdot 4^x = 0$. [3%]

b) If repetition is not allowed, how many five-digits even number smaller than 50000 can be formed by using 2,4,5,6,7,8,9? [3%]

8. a) Prove that $\dfrac{\cos(A-45^\circ)\sin(45^\circ+A)+\sin(A-45^\circ)\cos(45^\circ+A)}{\sin A}=2\cos A$ [3%]

b) Given that $f(x)=\log(3-x)$ and $g(x)=\sqrt{4-x}$. If $f(g(x))=0$, find x. [3%]

9. a) Find the general solution of the differential equation. [3%]

$$x\frac{dy}{dx}+2y=4x^2$$

b) Find the particular solution for which $y=5,\ x=1$, giving your answer in form of $y=f(x)$. [3%]

10. a) Find $\int \cos^3 2x \sin^3 2x \, dx$. [3%]

b) Given the focus of a parabola is (3, -4), the intersection point of directrix and axis of symmetry is (-1,-4), find the equation of the parabola. [3%]

11. a) Find the extreme point(s) of the curve $y=\dfrac{1}{2}e^{2x}-8e^x+12x$ and determine whether they are local maximum point of local minimum point. [4%]

b) Given that $12\sin A-5\cos B=9,\ 12\cos A+5\sin B=2$, find the value of $\sin(A-B)$. [3%]

c) Find $\lim\limits_{x\to -1}\left(\dfrac{1}{x+1}+\dfrac{x^2-3x-1}{x^2-x-2}\right)$ [3%]

12. a) Using mathematical induction, prove that for any positive integer n, [5%]

$$\frac{1}{3\times5\times7}+\frac{1}{5\times7\times9}+\ldots+\frac{1}{(2n+1)(2n+3)(2n+5)}=\frac{n(n+4)}{15(2n+3)(2n+5)}$$

b) Solve inequality $\dfrac{x^3-8x^2+7x}{4-3x-x^2}>1$ [5%]

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